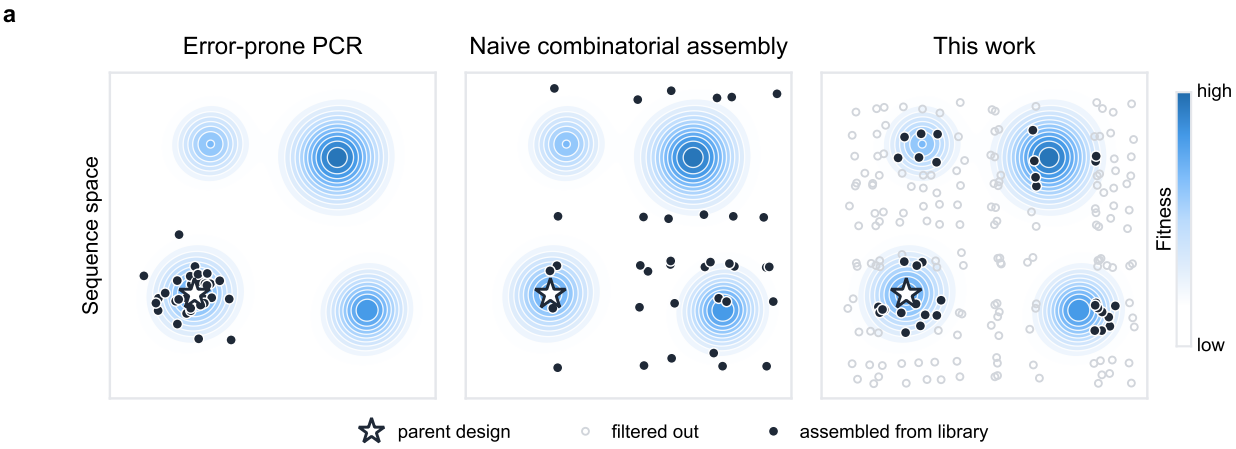
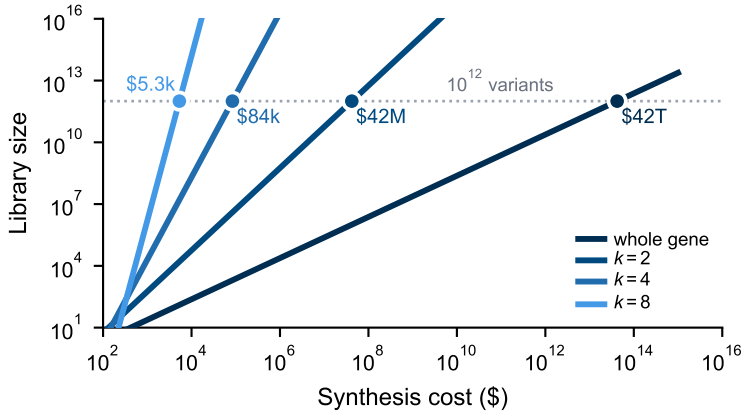
Design of high fidelity combinatorial libraries

November 5, 2026

Abstract

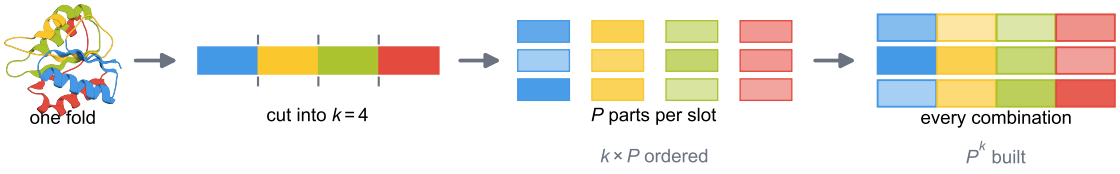
Recent tools for de novo design have revolutionized the way we propose novel proteins and the types of challenges we are able to tackle. While these tools have dramatically increased the hit rate of functional designs, zero-shot generations often require further optimization to enhance the desired properties. We present a novel method for generating high fidelity diverse combinatorial libraries. Traditional library generation methods rely on error prone PCR or natural fragment shuffling to generate diversity. Such approaches do not take into account the filtering of designs through structure prediction models or other desired oracles. We demonstrate the ability to post-train inverse folding models to design fragments with filter pass rates of 93.4% and libraries which can easily scale beyond 10^{12} individually addressable variants. Our post-training framework enables control over the search space by centering the library around a reference sequence with tunable mutation distance. Once designed, we demonstrate how individual variants are assembled with simple molecular biology techniques (golden gate assembly). Our cell-free assay platform enables us to screen enzymes at the scale of 10^3 measurements per day, each with replicates. The libraries generated are easily much larger than most traditional and modern high throughput assays are able to screen, so we demonstrate how the library can be sub sampled smartly to optimize enzymes. For a de novo designed PETase we demonstrate our platform finds a variant with 250x increase in catalytic activity.





c

panel (c) reserved



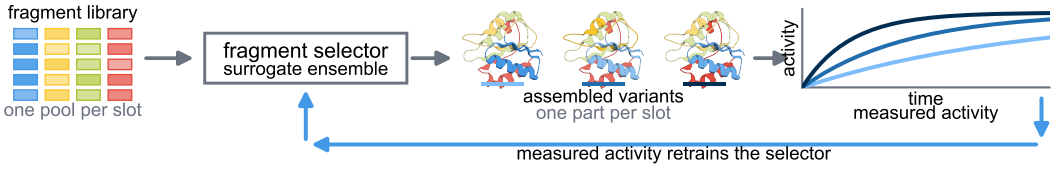


Figure 1: **A combinatorial library, and the loop around it.** [PLACEHOLDER — the prose is yours. Kept short so the float fits; the facts below are the ones a reader cannot get from the picture.] **(a)** One toy landscape searched three ways with the same number of variants built and screened ($N = 36$). Illustrative: no axis is measured. **(b)** eBlocks are billed at a 300-bp minimum, which binds from $k = 2$ on, so past that point cutting finer stops buying shorter DNA and only buys more pieces of it. 10^{12} variants costs \$42T as whole genes, \$42M at $k = 2$, \$84k at $k = 4$, \$5.3k at $k = 8$. **(c)** [Reserved — to be filled in.] **(d)** What makes the library combinatorial. One fold is cut at $k - 1$ boundaries; each slot is then an independent pool of P parts; and any one part per slot assembles a full-length design. Ordering $k \times P$ fragments therefore builds P^k sequences, which is the exponent panel (b) prices. Parts per slot are drawn as three to keep the columns legible. **(e)** The experimental loop. The arrow returns to the *selector*, not the library: parts are synthesised once and each round changes only which of them get combined.

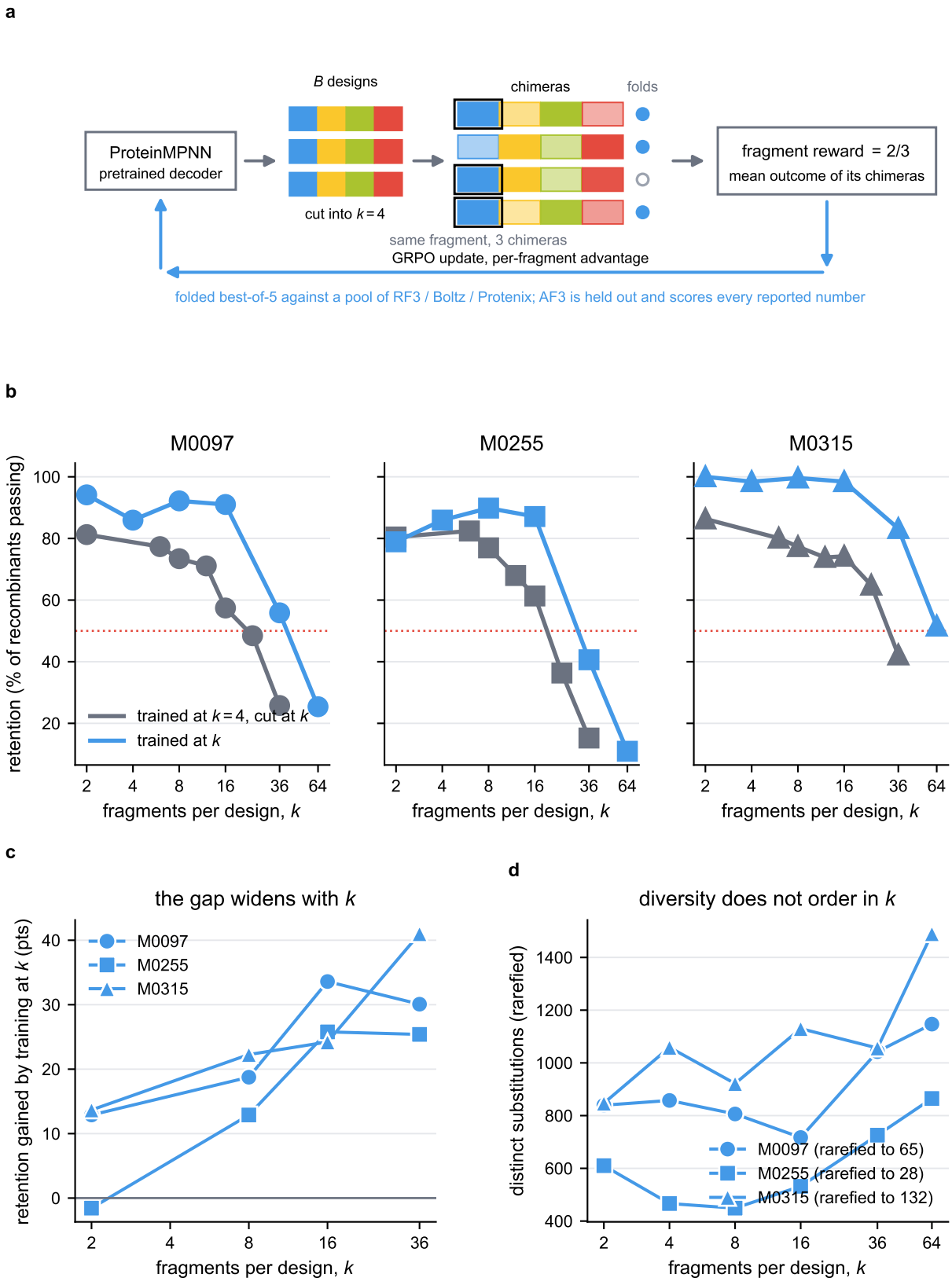

a
B designs
chimeras
folds
ProteinMPNN
pretrained decoder
cut into k = 4
fragment reward = 2/3
mean outcome of its chimeras
same fragment, 3 chimeras
GRPO update, per-fragment advantage
folded best-of-5 against a pool of RF3 / Boltz / Protenix; AF3 is held out and scores every reported number
b
M0097
M0255
M0315
retention (% of recombinants passing)
trained at k = 4, cut at k
trained at k
fragments per design, k
c
the gap widens with k
retention gained by training at k (pts)
M0097
M0255
M0315
fragments per design, k
d
diversity does not order in k
distinct substitutions (rarefied)
M0097 (rarefied to 65)
M0255 (rarefied to 28)
M0315 (rarefied to 132)
fragments per design, k

Figure 2: **Designing high-fidelity combinatorial libraries.** [PLACEHOLDER — the prose is yours.] **(a)** The method. A backbone conditions the ProteinMPNN decoder, treated as a policy; the sequences it emits are cut at the fragment boundaries, recombined across parents, and folded best-of-5 against a *pool* of oracles (RF3, Boltz, Protenix) under a round robin. Each fragment is scored by the recombinants it appears in, and that per-fragment reward drives the GRPO update. AlphaFold3 is held out of training entirely — it is probed every 25 steps as a diagnostic, never as reward — so every number in (b)–(d) is scored by a predictor the policy never optimised against. **(b)** Retention, the share of recombinants that still pass, against the number of fragments per design. Blue: a policy trained at that k. Grey: one policy trained at $k = 4$ and re-cut at k post hoc. Three backbones, six trained k levels, every trained cell at a matched 200 steps so training budget cannot covary with k. **(c)** The gap between the two, which widens monotonically with k — from +8.3 points at $k = 2$ to +32.2 at $k = 36$. **(d)** Rarefied substitution diversity over surviving fragments. It does not order in k, which is what rules out the alternative explanation for (b): a policy can satisfy "make recombinable fragments" by emitting near-identical sequences, and retention alone cannot tell that apart from genuine modularity. Caveats that must survive into the text: every cell is single-seeded against a documented 21-point run-to-run swing; M0255 at $k = 36$ and $k = 64$ was still climbing at step 200, so those are lower bounds.